

# **HTCS 6707**

## **Cybersecurity Project B**

### **Report**

By Luis Suarez (1600140), Fergus McKenzie () & Young Kwang Kweon  
(1605812)

# Table of Contents

---

|                                                                            |           |
|----------------------------------------------------------------------------|-----------|
| <b><a href="#">Introduction</a></b> .....                                  | <b>3</b>  |
| <b><a href="#">What is Cybersecurity Operation Centre (SOC)?</a></b> ..... | <b>4</b>  |
| <b><a href="#">Essential tools for SOC</a></b> .....                       | <b>5</b>  |
| <b><a href="#">Skills of Security Analyst</a></b> .....                    | <b>7</b>  |
| <b><a href="#">The Network Simulation</a></b> .....                        | <b>8</b>  |
| <a href="#">Technical Configurations</a> .....                             | 8         |
| <a href="#">Kali Linux</a> .....                                           | 10        |
| <a href="#">PFSense Firewall</a> .....                                     | 11        |
| <a href="#">Metasploitable</a> .....                                       | 13        |
| <a href="#">CentOS and Splunk Enterprise</a> .....                         | 14        |
| <a href="#">Windows Server 2019 and Splunk Universal Forwarder</a> .....   | 20        |
| <a href="#">Security Mechanism</a> .....                                   | 27        |
| <a href="#">Red Team</a> .....                                             | 27        |
| <a href="#">Blue Team</a> .....                                            | 32        |
| <b><a href="#">Conclusion</a></b> .....                                    | <b>36</b> |
| <b><a href="#">Resources and References</a></b> .....                      | <b>37</b> |

# Introduction

---

The purpose of this project is to guide readers on the fundamentals of the Cybersecurity Operations Centre (SOC) and demonstrate the design, implementation, and evaluation of a simulated SOC within a controlled, virtualised environment.

This report covers what the Security Operations Centre is, essential tools for SOC, skills required to become a successful Security Analyst, and technical configurations and security mechanisms of our virtual SOC simulation.

By constructing a small enterprise-style network and conducting both Red Team and Blue Team testing, our project aimed to demonstrate the core functions of a SOC: monitoring, detecting, analysing, and responding to security events. Our report documents the network configuration, the security mechanisms implemented, and the behaviours of key components, including the Windows Server 2019 Domain Controller, the PFSense Firewall, the Splunk Enterprise SIEM, and the simulated attacker and target machines.

# What is Cybersecurity Operation Centre (SOC)?

---

A Security Operations Centre (SOC) is a team of analysts, primarily cybersecurity experts, whose purpose is to prevent, detect, assess, respond to, and report on security incidents.

A SOC can have various functions, such as gathering data on cyber threats through data analysis from multiple sources, including reports, raw data, and external sources, or even developing its own tools. While not all SOC operations are the same, all those mentioned above share the ultimate goal of identifying and responding to intrusions.

SOCs can operate with small teams of analysts, where each team member is technically expected to "do everything." There are also more specialised SOC, such as national coordination centres, which have larger teams of analysts and response personnel, each with a single, specific task.

The SOC missions include:

- Continuous threat analysis
- Vulnerability scanning
- Implementation of coordinated countermeasures
- Advising on security policies and architecture
- Monitoring, detecting, and analysing potential intrusions in real time and by searching for adversaries, using various security-relevant data sources
- Responding to confirmed incidents by coordinating resources and applying timely and appropriate countermeasures
- Providing situational information and reports on the state of cybersecurity, incidents, and trends in adversary behaviour to relevant organizations
- Engineering and operating SOC technologies, such as host sensors, network sensors, and log collection and analysis systems

The SOC's mission can also be conceived in terms of the infrastructure and data it protects. For some smaller SOC, this may consist solely of on-premises digital assets and data. However, for most SOC, their responsibilities also include monitoring, detecting, and responding to potential incidents in remote systems, cloud systems, data, and community mobile infrastructure.

# Essential tools for SOC

---

A Security Operations Centre (SOC) relies on a set of tools that enable rapid and coordinated detection, analysis, and response to security incidents. While each organisation may have variations, specific components are essential for a SOC to function efficiently and maintain good visibility into what is happening within the network.

A critical cybersecurity tool is **SIEM (Security Information and Event Management)**. The purpose of this tool is to compile information from logs originating from multiple sources within an enterprise's environment and alert analysts to any suspicious activity. The advantage of using SIEMs like Splunk or Elastic SIEM is that analysts can identify patterns that could not have been observed by separate review of each system. The SOC is literally "blind" without a working SIEM.

The other crucial skill is provided by **IDS/IPS (Intrusion Detection & Prevention) Systems**. These programs examine network activity and determine any indication of an attack being perpetrated, including port scans, payload activity, and exploit attempts. Although an intrusion detection system merely notifies, an intrusion prevention system can actually cut off an attacking connection. This is particularly crucial to have programs such as Snort or Suricata to alert on approaching attacks to sensitive computers.

Firewalls, particularly **Next-Generation Firewalls (NGFWs)**, are also essential within a SOC environment. They assist in controlling incoming and outgoing traffic, while most contemporary firewalls can actually inspect packets and filter malicious information. For many organisations, firewalls represent one of the initial points where suspicious activity is identified.

Another tool with significant activity is **Endpoint Protection**, making SOC teams highly dependent on **EDR/XDR (Endpoint Detection and Response or Extended Detection and Response)** solutions. These solutions operate on employees' devices and servers, monitoring unusual processes, malware, and lateral activity within a network. This type of product can decrease the probability that an infected device remains undetected, utilising something like Microsoft Defender Endpoint or CrowdStrike.

Finally, having a SOC is aided by **vulnerability scanners**, such as Nessus or Qualys, which allow one to discover vulnerabilities prior to hackers discovering them. Then, when an

incident occurs, aiding incident response through ticketing and case management applications such as ServiceNow or TheHive can be very useful.

These technologies, when implemented simultaneously, provide a SOC with the capability to address any security threat that may arise.

# Skills of Security Analyst

---

To become a successful security analyst, they need a solid mix of complex and analytical skills, based on a solid understanding of networking technology (TCP/IP, OSI model) as well as operating systems (Windows, Linux), to identify anomalies in network traffic and running systems.

The foundational knowledge of using tools like SIEM and EDR to log events and monitor threats is also required, and the analyst needs to be able to carry out the incident response lifecycle (triage, containment, eradication, recovery).

Technical skills alone will not suffice in this role; the analyst will be challenged with many situations involving determining false positives in an incident versus a real attack, so they need to be able to think critically and can communicate well to express complex risks to non-technical stakeholders.

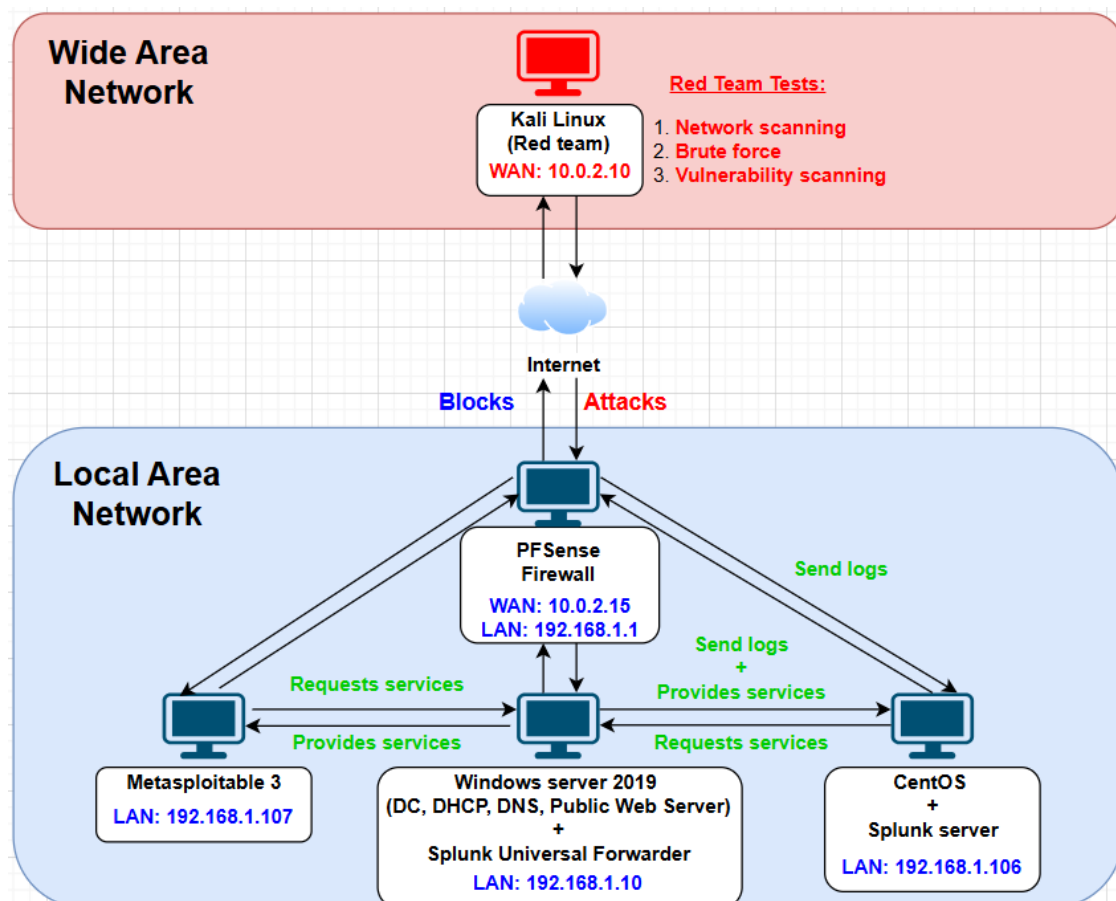
# The Network Simulation

## Technical Configurations

This section covers the simulation of the organisational Security Operation Centre. To simulate SOC securely, we conducted our simulation lab in a virtual environment using VirtualBox to create five virtual machines, install and run all operating systems. Figure 1 below illustrates the structure of our SOC simulation:

**Figure 1**

*Network diagram*



Note. This network diagram for SOC simulation environment was generated using <https://app.diagrams.net/>

We used five virtual machines to set up our project:

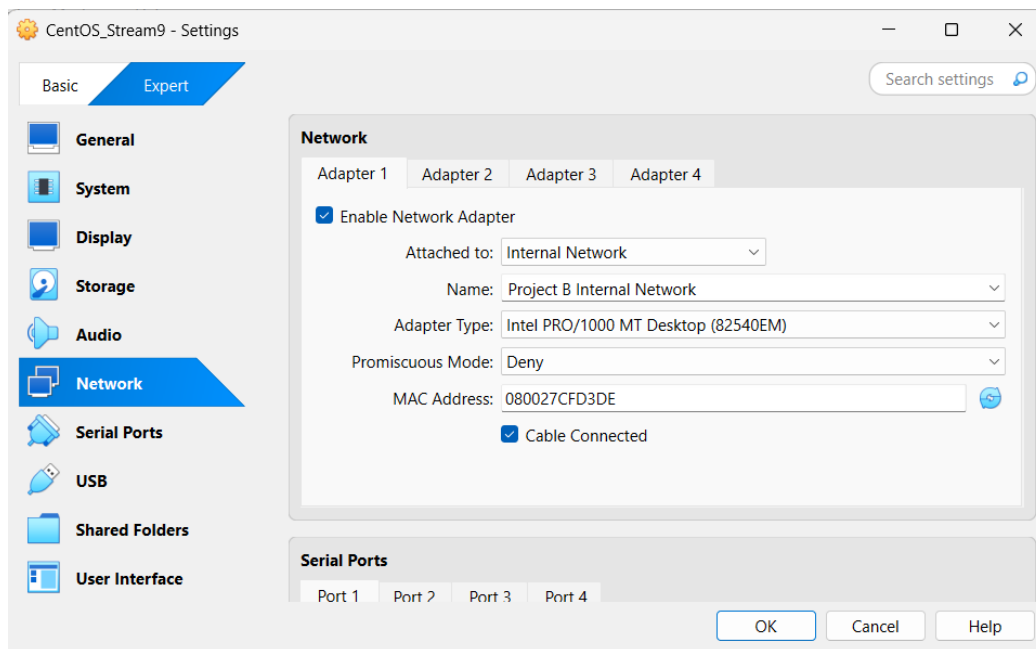


- **Kali Linux:** To conduct three Red Team Tests against the local network from outside.
- **PFSense Firewall:** A Firewall and a router to protect the local network from attacks from outside the network and send logs to the Splunk server.
- **Metasploitable:** To test the network's vulnerability.
- **CentOS with Splunk Enterprise installed:** For SIEM simulation.
- **Windows Server 2019 with Splunk Universal Forwarder installed:** To provide services to other hosts and simulate log collection and forwarding to Splunk.

**PFSense Firewall, Metasploitable, CentOS, and Windows Server 2019** have their network adapters connected to the same internal network called '**Project B Internal Network**' and can communicate with each other. Refer to Figure 2 below:

**Figure 2**

*Network Configuration of CentOS virtual machine*

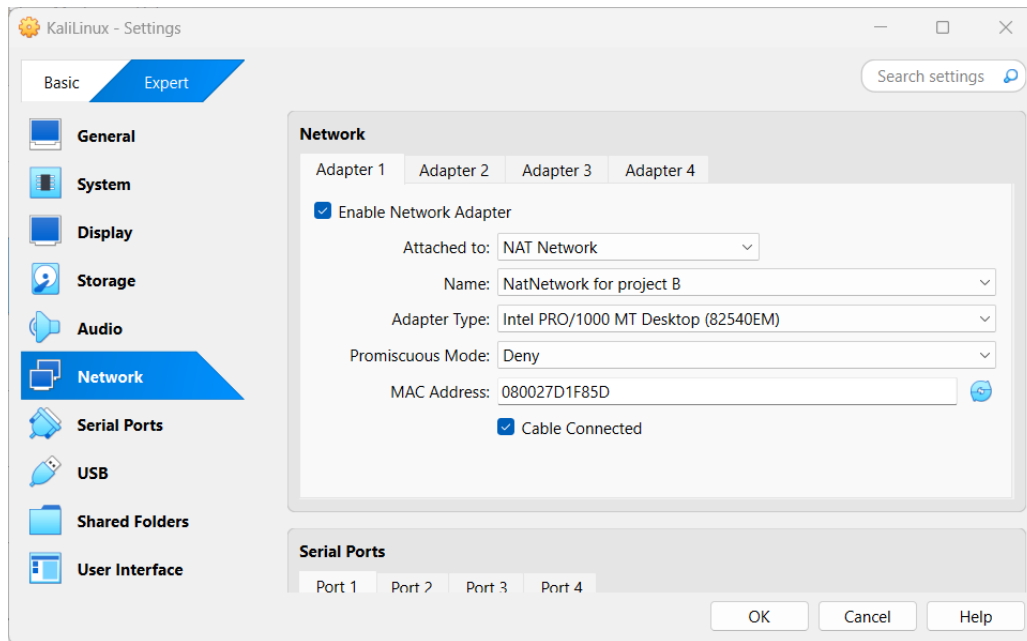


*Note.* This Figure demonstrates the network setup for internal networks in VirtualBox

**PFSense** has another network adapter called NAT network connected to '**NATNetwork for project B**' for internet access. **Kali Linux** is also connected to this NAT Network, allowing it to attempt to access the internal network by passing through PFSense. Refer to Figure 3 below:

**Figure 3**

*Network Configuration of Kali Linux virtual machine*



*Note.* This Figure demonstrates the network setup for NAT network in VirtualBox

Below are the technical configurations of each machine:

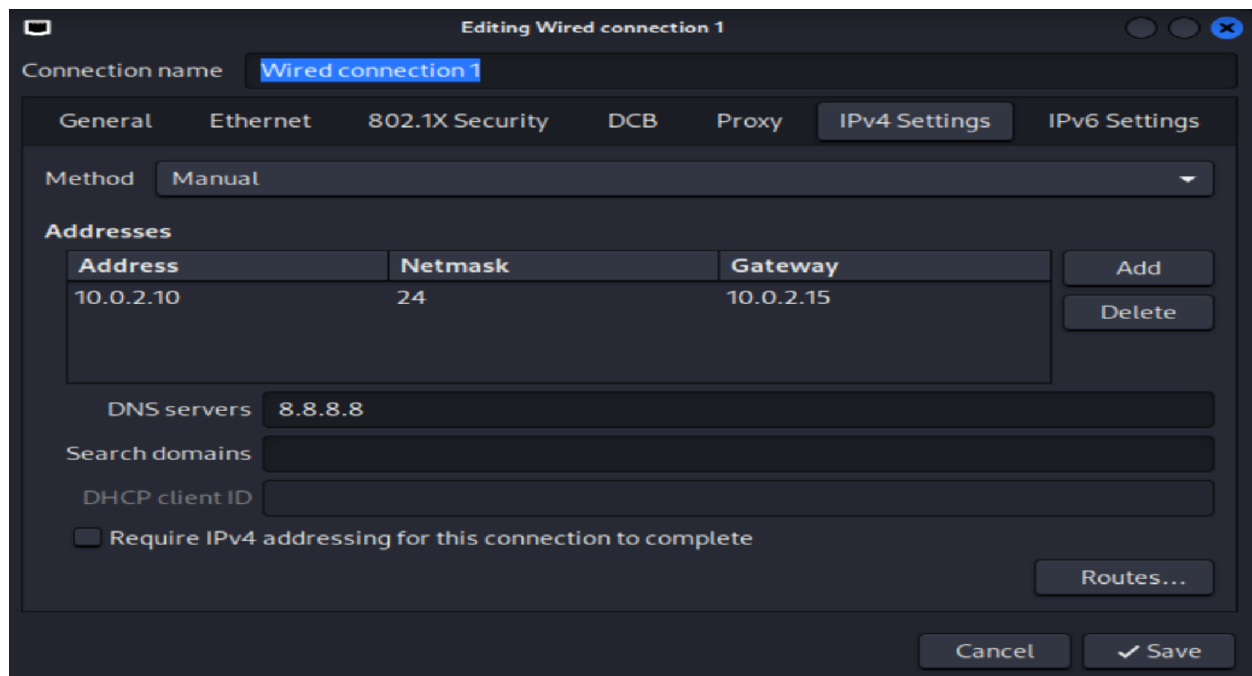
## Kali Linux

Kali Linux is a Debian-based Linux distribution for penetration testing. It contains several tools and scripts to allow users to simulate real-world cyberattacks. It served as our Red Team attack machine to conduct the three tests, which will be discussed in the '**Security Mechanism**' section.

In our simulation, we assigned '**10.0.2.10**' as Kali's static NAT IP address. We set '**10.0.2.15**' as the default gateway, which is the NAT IP address of the PFSense firewall, so that we can attempt to pass through the firewall and carry out attacks against Metasploitable, which is in the local network. By default, Kali cannot carry out attacks or ping any hosts within the local network, as PFSense is configured to block all traffic from outside the local network.

**Figure 4**

*IP Configuration of Kali Linux*



Note. This Figure demonstrates the IP set up of Kali with 'Advanced Network Configuration'

## PFSense Firewall

PFSense is an open-source firewall and router operating system. It offers many network services, including intrusion detection and prevention, and DNS filtering. The firewall's DHCP service was disabled, allowing the Windows Server 2019 to function as the sole DHCP server for our network.

In our local network, it was configured as the main perimeter firewall responsible for routing and traffic control between the WAN and LAN networks. It was deployed with the two static interfaces to ensure stable routing between the two network segments.

### **Figure 5**

*PFSense Firewall interface*

```
PFSense [Running] - Oracle VirtualBox
Starting syslog...done.
Starting CRON... done.
pfSense 2.7.2-RELEASE amd64 20231206-2010
Bootup complete

FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)

VirtualBox Virtual Machine - Netgate Device ID: b807f6efffae50d7bb22

*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***

WAN (wan)      -> em0      -> v4: 10.0.2.15/24
LAN (lan)      -> em1      -> v4: 192.168.1.1/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults  13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                 15) Restore recent configuration
7) Ping host                   16) Restart PHP-FPM
8) Shell

Enter an option: █
```

Note. This Figure demonstrates IP configuration of PFSense firewall

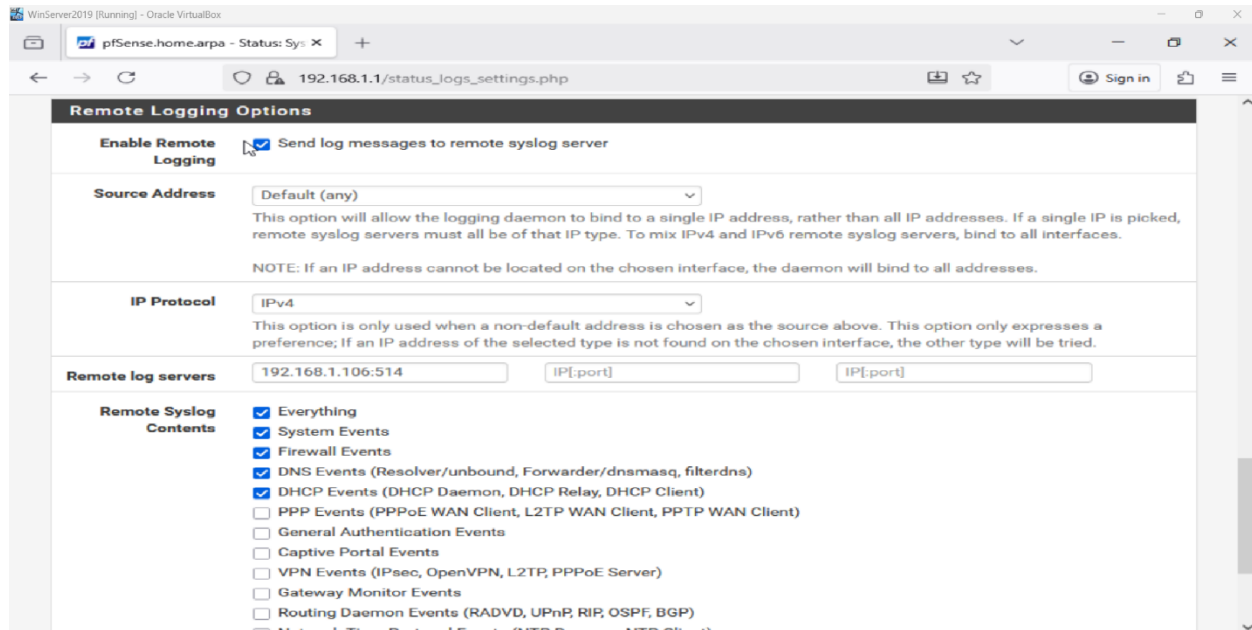
‘10.0.2.15’, the WAN interface serves as a gateway for all traffic from outside the local network and blocks all inbound connections to internal devices unless explicitly permitted. ‘192.168.1.1’, the LAN interface serves as the default gateway for all hosts within the local network, allowing outbound traffic from them to reach the internet or external networks.

It senses all traffic activities inside and outside the local network and sends all activity logs to CentOS, where Splunk Enterprise is installed. By default, it does not send logs to Splunk automatically, so configurations from both Splunk and the PFSense web interface are required. This can be done by:

1. Log in to the PFSense web interface either from **CentOS** or **Windows Server 2019**
2. Go to **Status → System Logs → Settings**
3. In ‘**Remote Logging Options**’, tick ‘**Send log messages to remote syslog server**’.
4. Enter CentOS IP address (**192.168.1.106**) and UDP port number (**514**)

**Figure 6**

*Configuration of PFSense Firewall for log forwarding*



Note. This Figure demonstrates configuration of log forwarding from PFSense's web interface

Configuration for Splunk Enterprise will be discussed in the '**CentOS and Splunk Enterprise**' section.

## Metasploitable

Metasploitable is an intentionally vulnerable Ubuntu Linux distribution that is designed for testing security vulnerabilities. The system contains outdated web applications, open ports, weak credentials, and known vulnerabilities, allowing the Red Team to perform exploitations. We used '**Metasploitable 3**', the latest version, to run our tests efficiently.

**Figure 7**

*Metasploitable 3 Interface*

```
metasploitable3-ub1404 login: vagrant
Password:
Last login: Sat Nov 15 03:25:10 UTC 2025 on tty1
Welcome to Ubuntu 14.04.6 LTS (GNU/Linux 3.13.0-170-generic x86_64)

 * Documentation:  https://help.ubuntu.com/
vagrant@metasploitable3-ub1404:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:e1:6b:89 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.107/24 brd 192.168.1.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe01:6b89/64 scope link
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 02:42:92:29:4e:89 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
    inet6 fe80::42:92ff:fe29:4e89/64 scope link
        valid_lft forever preferred_lft forever
vagrant@metasploitable3-ub1404:~$
```

Note. This figure demonstrates IP configuration of Metasploitable 3

Metasploitable itself does not have an IP address, so it requires a DHCP server to assign one to it. In our local network, Windows Server 2019 serves as the DHCP server for the network and assigns the IP address ‘192.168.1.107’ to Metasploitable.

## CentOS and Splunk Enterprise

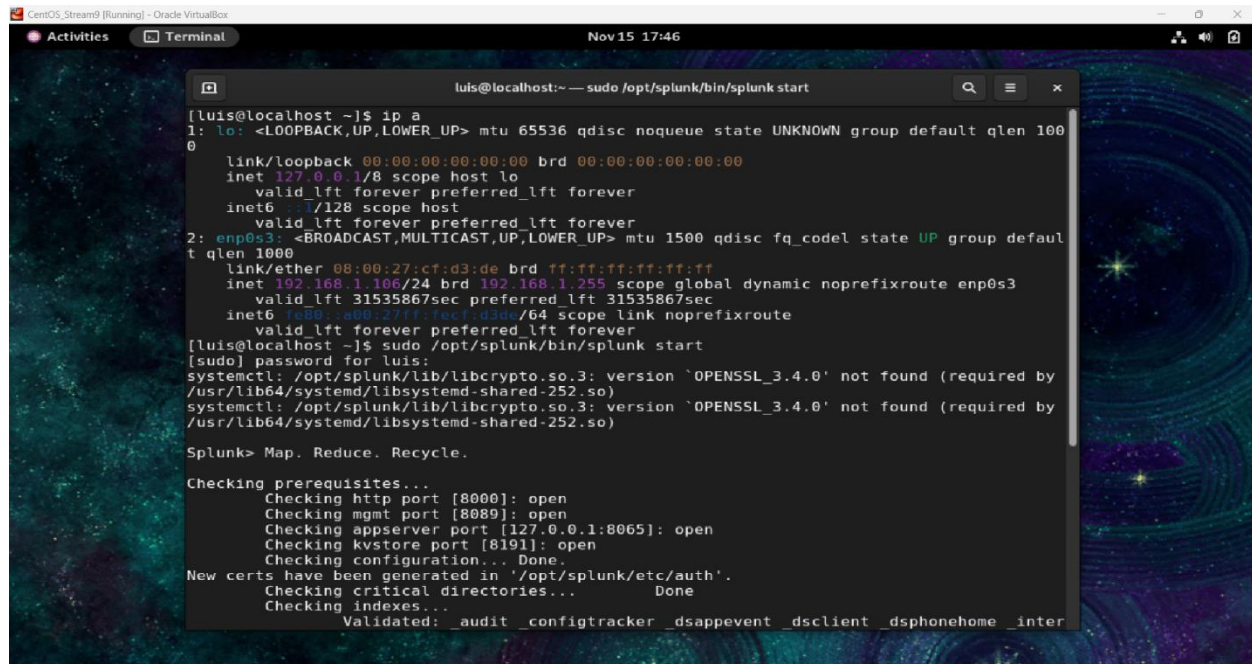
CentOS is an open-source Linux distribution widely used in industries for web hosting, database hosting, application servers, file servers, FTP servers, proxy servers, and many other purposes, thanks to its excellent security features (Liquid Web, 2019).

Splunk Enterprise is a popular SIEM tool for log collection, analysis, searching, and visualisation. In our local network, it was installed on CentOS for enhanced security and acts as a core SIEM, receiving logs from PFSense and Windows Server 2019.

Like Metasploitable, CentOS itself does not have an IP address, so it requires a DHCP server, such as **Windows Server 2019**, to distribute the IP address. Without the IP address, it would be unable to communicate with other hosts in the network. By default, CentOS does not have a DHCP client installed. To install the DHCP client, go to terminal -> type ‘**sudo dnf install dhclient**’. After installation, the DHCP server from Windows Server will automatically assign an IP address to CentOS. The IP address assigned to its network interface, ‘**enp0s3**’ from Windows Server, is ‘**192.168.1.106**’ as shown in Figure 8 below:

**Figure 8**

## CentOS Network interface



The screenshot shows a terminal window titled "CentOS\_Stream9 [Running] - Oracle VirtualBox" with a date and time of "Nov 15 17:46". The terminal displays the output of the command `luis@localhost:~$ ip a`, showing details for the loopback interface `lo` and the ethernet interface `enp0s3`. Following this, the command `luis@localhost:~$ sudo /opt/splunk/bin/splunk start` is executed. The terminal shows a password prompt for `luis`, followed by systemctl status messages for `/opt/splunk/lib/libcrypto.so.3`. The Splunk startup process then begins, displaying "Map. Reduce. Recycle." and "Checking prerequisites...". It checks the status of ports 8000, 8080, 8089, and 8191, all of which are open. It also checks the configuration and generates new certificates in `/opt/splunk/etc/auth/`. Finally, it checks critical directories and indexes, listing validated components: `_audit _configtracker _dsappevent _dsclient _dsphonehome _inter`.

```
[luis@localhost ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
    t qlen 1000
    link/ether 08:00:27:cf:d3:de brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.106/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid lft 31535867sec preferred_lft 31535867sec
    inet6 fe80::a00:27ff:feef:d3de/64 scope link noprefixroute
        valid lft forever preferred_lft forever
[luis@localhost ~]$ sudo /opt/splunk/bin/splunk start
[sudo] password for luis:
systemctl: /opt/splunk/lib/libcrypto.so.3: version 'OPENSSL_3.4.0' not found (required by
/usr/lib64/systemd/libsystemd-shared-252.so)
systemctl: /opt/splunk/lib/libcrypto.so.3: version 'OPENSSL_3.4.0' not found (required by
/usr/lib64/systemd/libsystemd-shared-252.so)

Splunk> Map. Reduce. Recycle.

Checking prerequisites...
  Checking http port [8000]: open
  Checking mgmt port [8089]: open
  Checking appserver port [127.0.0.1:8065]: open
  Checking kvstore port [8191]: open
  Checking configuration... Done.
New certs have been generated in '/opt/splunk/etc/auth'.
  Checking critical directories... Done
  Checking indexes...
    Validated: _audit _configtracker _dsappevent _dsclient _dsphonehome _inter
```

Note. This Figure demonstrates IP configuration of CentOS and script to run Splunk Enterprise

Once Splunk Enterprise is installed, we need a terminal and run script to open it in the browser. As shown in the Figure 8, the script to run is:

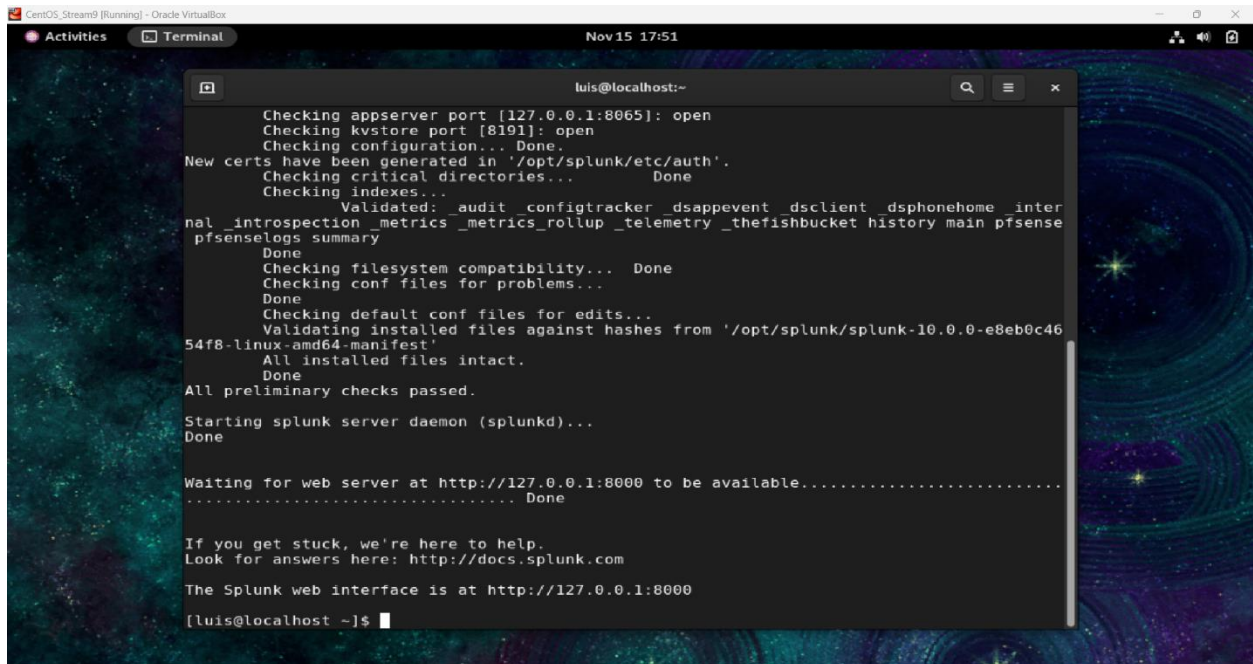
**`/sudo /opt/splunk/bin/splunk start`**

Once it started successfully, it shows the IP address of Splunk web interface as shown in Figure 9 below:

**Figure 9**

*Running Splunk Enterprise*





```
CentOS_Stream9 [Running] - Oracle VirtualBox
Nov 15 17:51

luis@localhost:~
Checking appserver port [127.0.0.1:8065]: open
Checking kvstore port [8191]: open
Checking configuration... Done.
New certs have been generated in '/opt/splunk/etc/auth'.
Checking critical directories... Done
Checking indexes...
Validated: audit configtracker dsappevent dsclient dsphonehome inter
nal_introspection_metrics_metrics_rollup_telemetry_thefishbucket history main pfsense
pfsenselogs summary
Done
Checking filesystem compatibility... Done
Checking conf files for problems...
Done
Checking default conf files for edits...
Validating installed files against hashes from '/opt/splunk/splunk-10.0.0-e8eb0c46
54f8-linux-amd64-manifest'
All installed files intact.
Done
All preliminary checks passed.
Starting splunk server daemon (splunkd)...
Done

Waiting for web server at http://127.0.0.1:8000 to be available.....
..... Done

If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com

The Splunk web interface is at http://127.0.0.1:8000

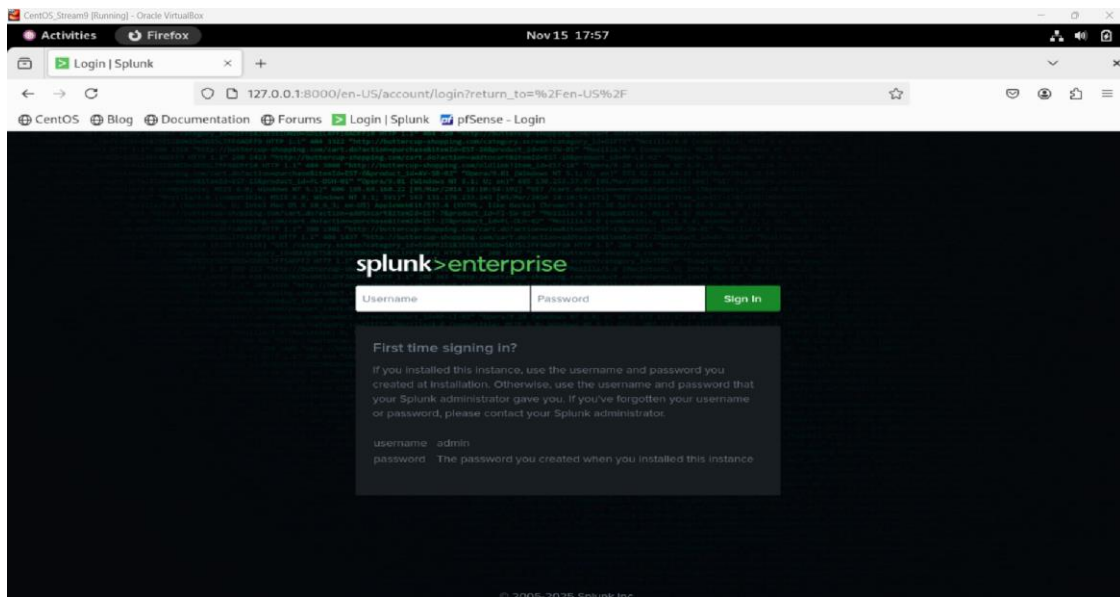
[luis@localhost ~]$
```

Note. This Figure demonstrates Splunk Enterprise is running

The Splunk web interface runs at '**http://127.0.0.1:8000**'. By copying and pasting it in the browser, it directs to Splunk Enterprise Web Interface login screen as shown in Figure 10:

**Figure 10**

*Splunk Enterprise Web Interface login*



Note. This Figure demonstrates Splunk Enterprise Web Interface is running successfully from CentOS

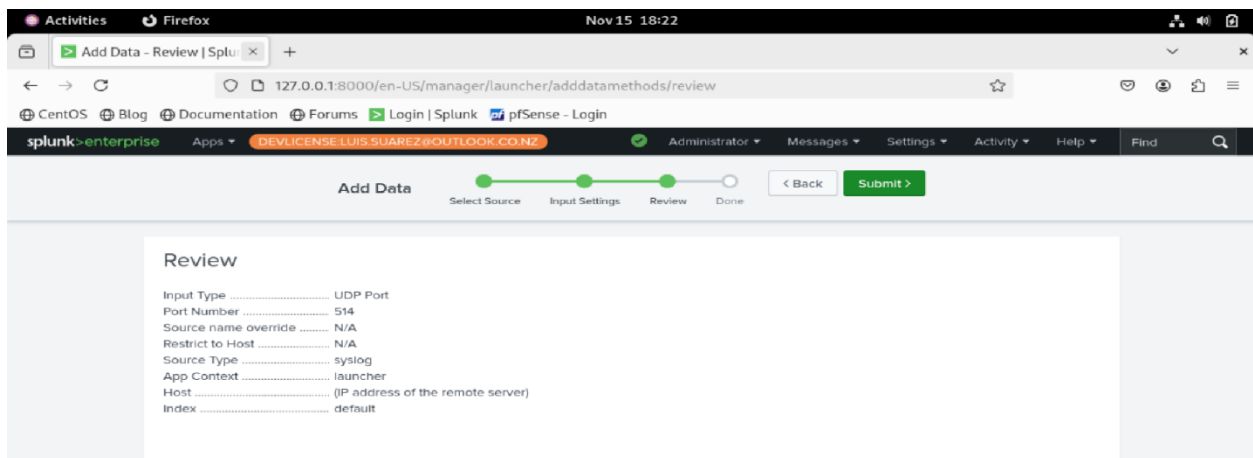


To receive logs from PFSense, some configurations and installation are required. In Splunk Enterprise server:

1. Log in to Splunk Web
2. Go to **Settings** → **Data Inputs** → **UDP** → **Add new**
3. Add 514 in 'port' section
4. Set the source type to 'Syslog'
5. Click **Review**, then **Submit**

**Figure 11**

*PFSense log configuration in Splunk Enterprise*



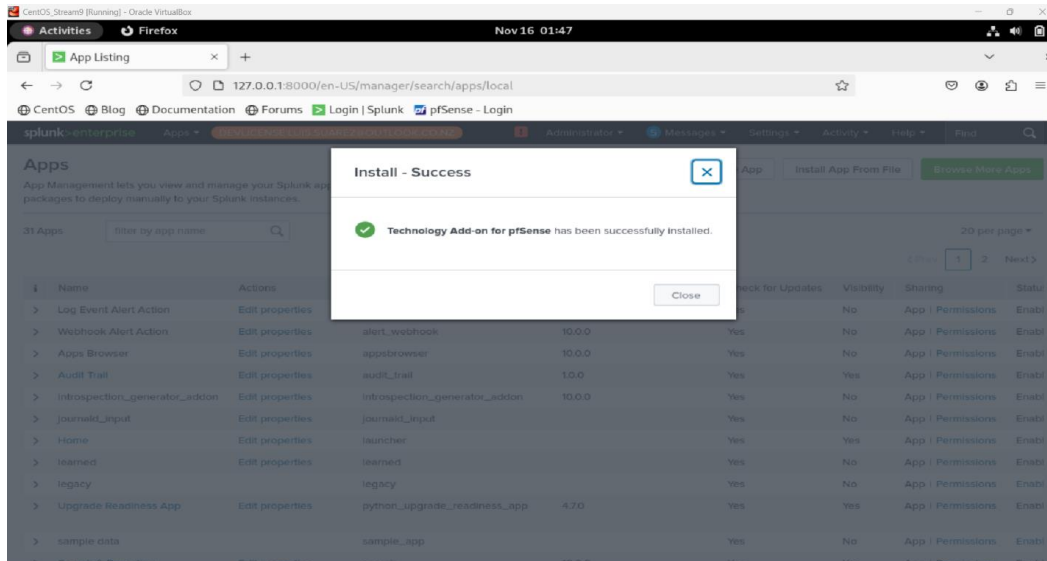
*Note.* This Figure demonstrates the configuration of UDP in Splunk Enterprise to receive logs from PFSense

'TA-pfsense' also needs be installed in Splunk Enterprise to receive these logs:

1. Download 'TA-pfsense' from Splunkbase on CentOS
2. Go to **Apps** > **Manage App** > **Install App from File**
3. Browse the downloaded TA-pfsense and click 'Upload'

**Figure 12**

*Installation of TA-pfsense*



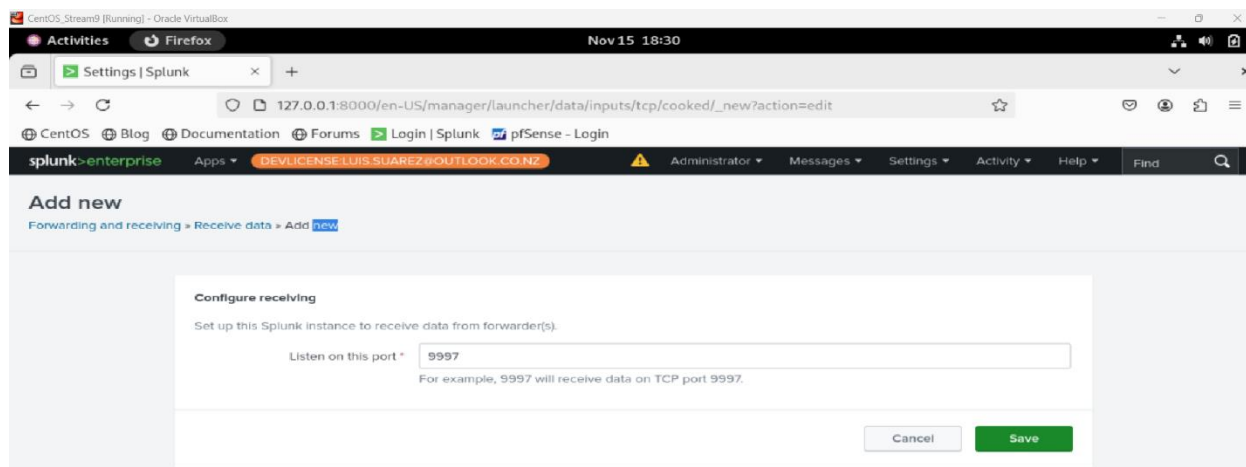
Note. This Figure demonstrates successful installation of TA-pfsense on Splunk Enterprise

This will allow Splunk to receive logs from PFSense. To receive logs from Windows Server, configurations are required in Splunk Enterprise, and **the Splunk Universal Forwarder must be installed on the Windows Server** to forward logs to Splunk Enterprise. This will be discussed in the ‘**Windows Server 2019**’ and ‘**Splunk Universal Forwarder**’ section. On Splunk Enterprise server:

1. Go to **Settings → Forwarding and receiving → Configure receiving → Add new**
2. Write ‘**9997**’ in the ‘**Listen on this port**’ section and **save**

**Figure 13**

*Windows Server log configuration in Splunk Enterprise*



Note. This Figure demonstrates the configuration of port 9997 in Splunk Enterprise to receive logs from Windows Server

After all the configurations in Splunk Enterprise are finished, from CentOS, open terminal and open its ports by using following scripts:

```
sudo firewall-cmd --permanent --add-port=514/udp
sudo firewall-cmd --permanent --add-port=8000/tcp
sudo firewall-cmd --permanent --add-port=8080/tcp
sudo firewall-cmd --permanent --add-port=8088/tcp
sudo firewall-cmd --permanent --add-port=8089/tcp
sudo firewall-cmd --permanent --add-port=9997/tcp
```

And then reload firewall:

```
sudo firewall-cmd --reload
```

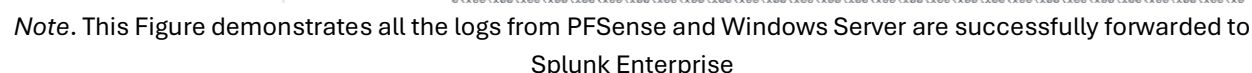
After successful configuration, logs from PFSense or Windows Server can be checked by:

1. From Splunk Enterprise home, go to 'search & Reporting'
2. Inside search box, type:  
**index=\* host="192.168.1.1" OR host="WindowsServer2019"** and press enter

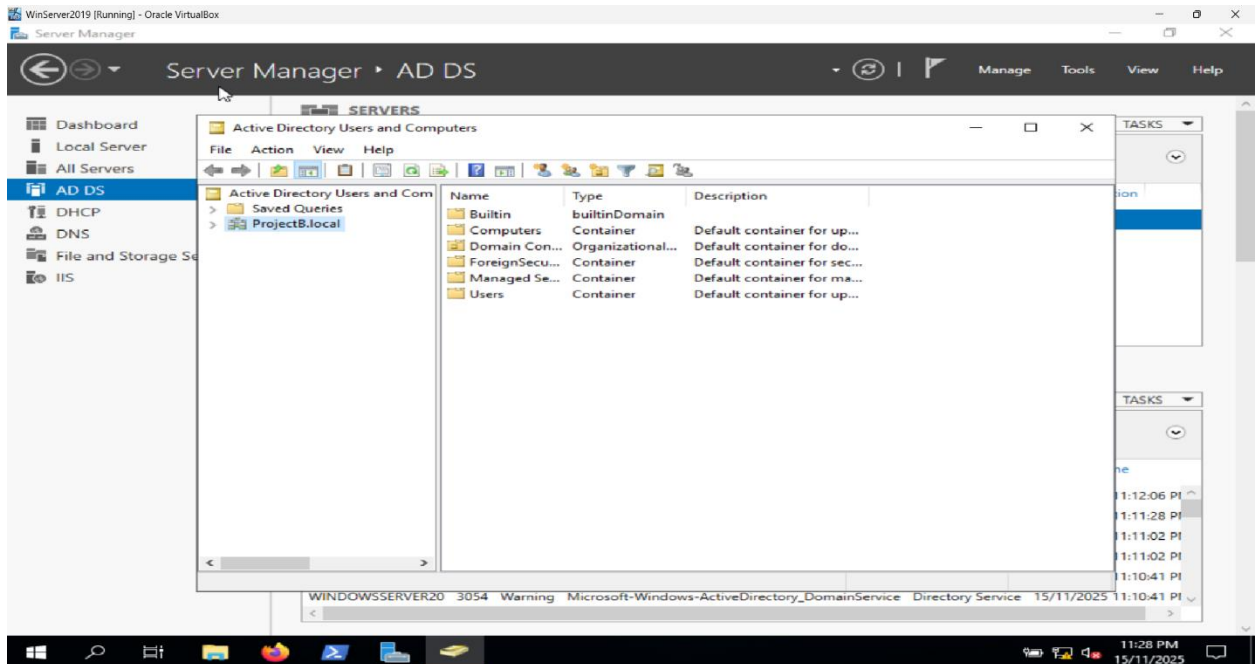
This will show all the logs received from '192.168.1.1' (PFSense's IP address), and 'WindowsServer2019' (host name of Windows server machine) as shown below:

## Figure 14

*Logs received from PFSense and Windows Server in Splunk Enterprise*

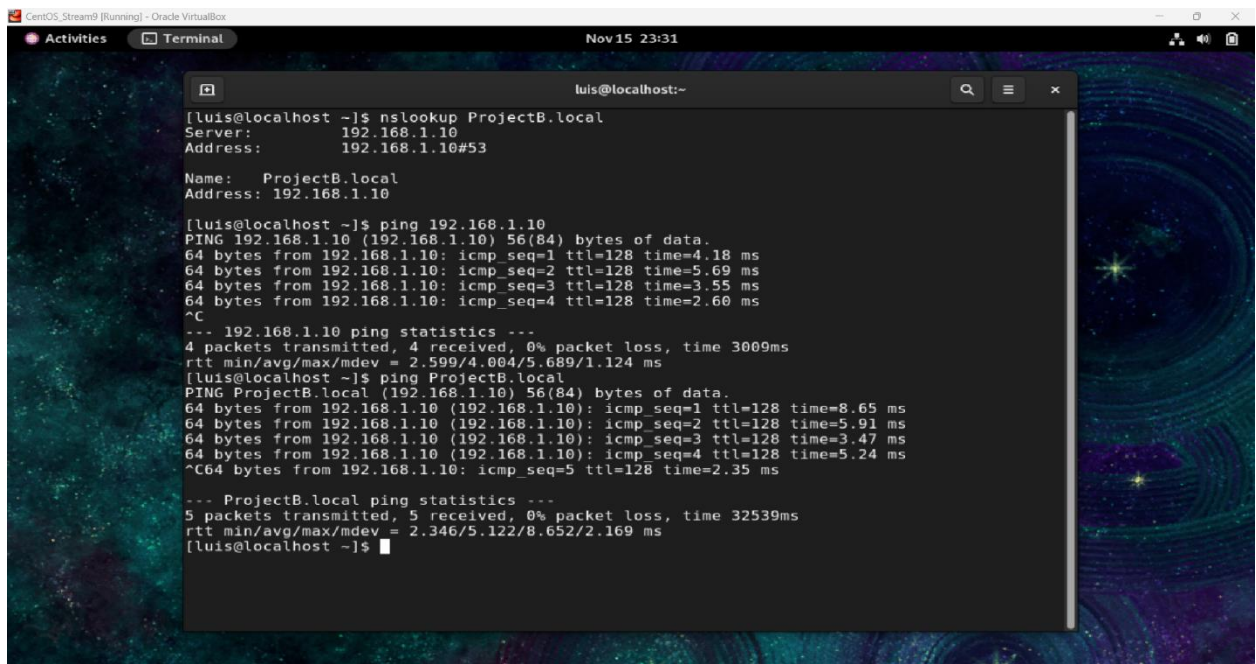


**Figure 15**  
*Domain Controller of Windows Server 2019*



Note. This Figure shows the name of domain server of Windows Server  
From CentOS, we can check DC in Windows Server using **nslookup** with its name:

**Figure 16**  
*Domain Controller checkups from Centos*

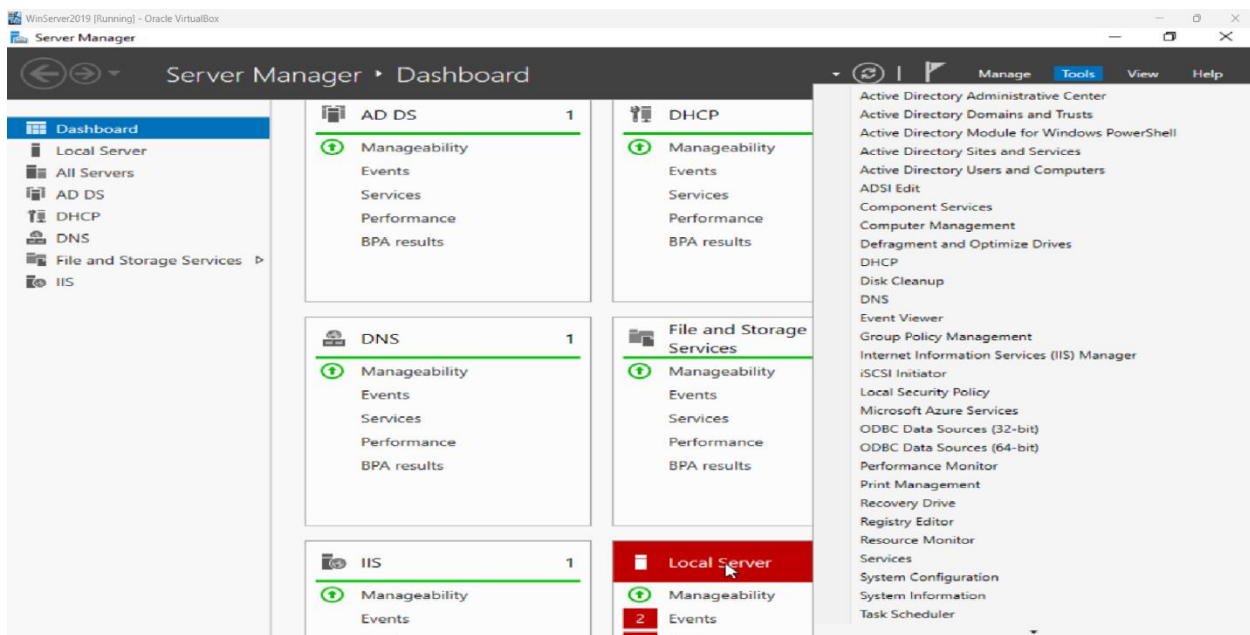


Note. This Figure demonstrates how to check Windows Server's DC is working from CentOS  
Figure 16 revealed that the Windows Server's IP address is '**192.168.1.10**' and can be pinged from CentOS.

- **DHCP Server:** Distributes IP addresses ranging from **192.168.1.100** to **192.168.1.200** to Metasploitable and CentOS.
- **DNS Server:** Translates human-readable domain names into IP addresses that computers use to communicate.
- **Public Web Server (IIS Web Server):** Allows Windows Server to host websites, web applications, APIs, and services.

**Figure 17**

*Server Manager in Windows Server 2019*



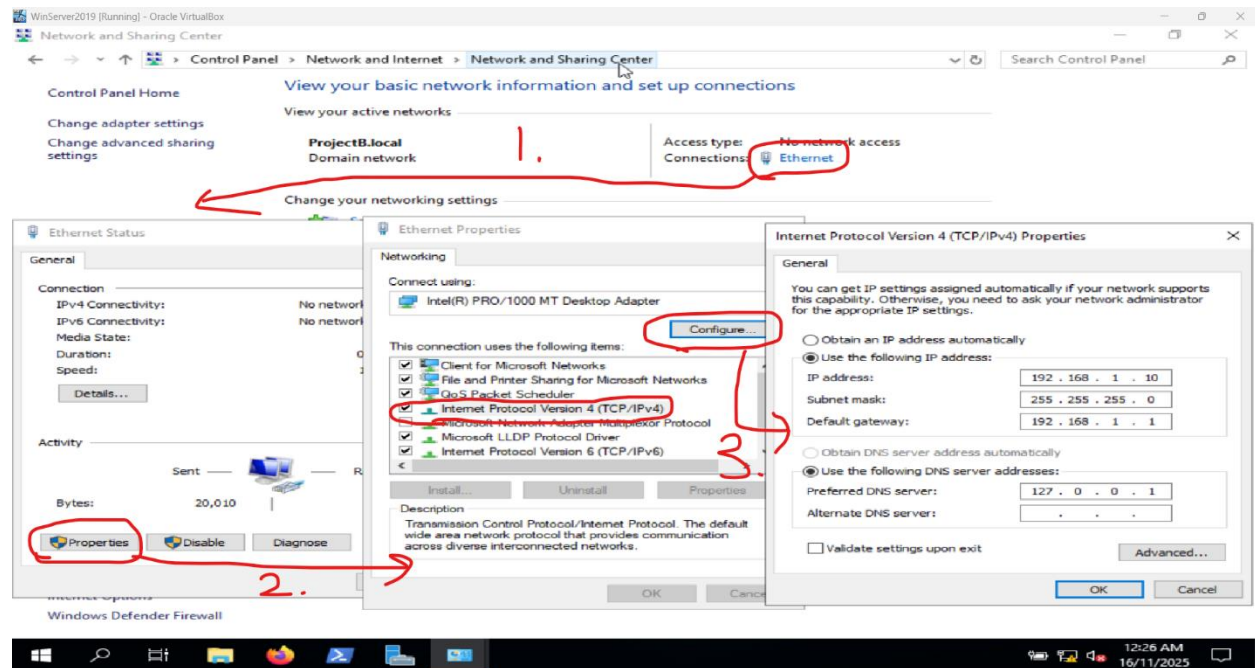
Note. This Figure demonstrates services in the Windows Server are running

For Windows Server to provide these services to other hosts, it also needs to have its own IP address and default gateway assigned. To do so:

1. Go to **Control Panel > Network and Internet > Network and Sharing Center**
2. Click **'Ethernet'** in **'View your active networks'** panel
3. Click **'Properties'**
4. Select **'Internet Protocol Version 4 (TCP/IPv4)'** and click **'Properties'**
5. Type IP address for Windows Server (we chose **'192.168.1.10'**) in **'IP address'** section and type PFSense's IP address (**192.168.1.1**) in **'Default gateway'** section and then click **'OK'**

**Figure 18**

*Step by step guide on configuration of Windows Server's network*



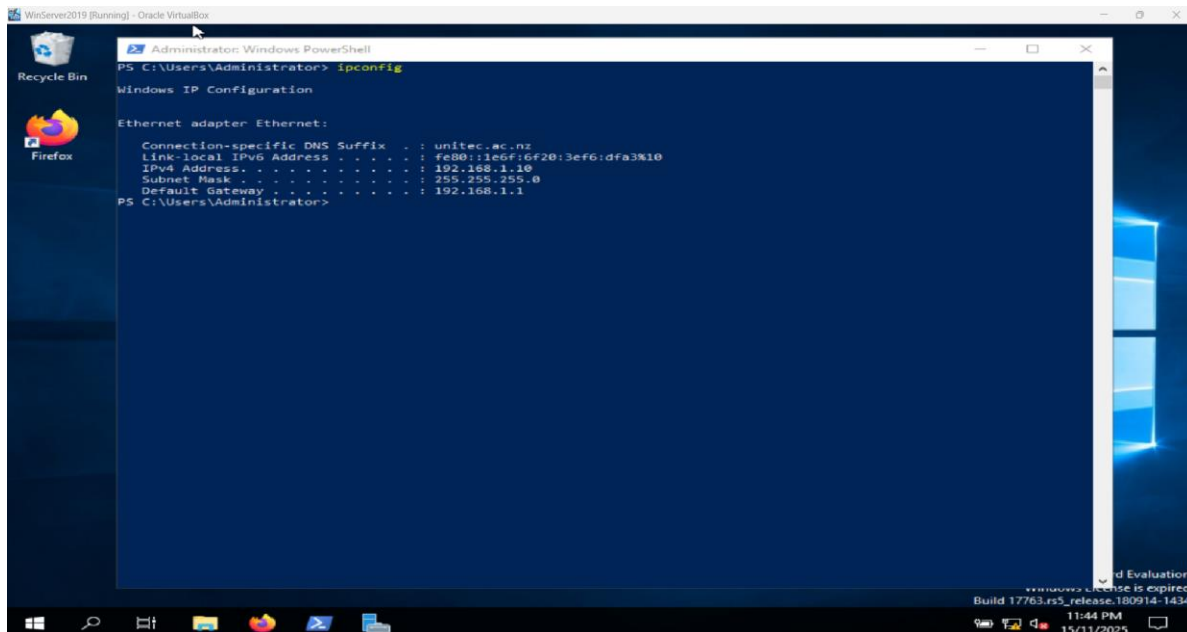
Note. This Figure demonstrates guides on how to assign static IP Address and default gateway of Windows Server

Once fully configured, we can check its network configuration in power shell using 'ipconfig' as shown in Figure 19 below:

**Figure 19**

*IP configuration of Windows Server 2019*





Note. This Figure demonstrates IP address and default gateway of Windows Server

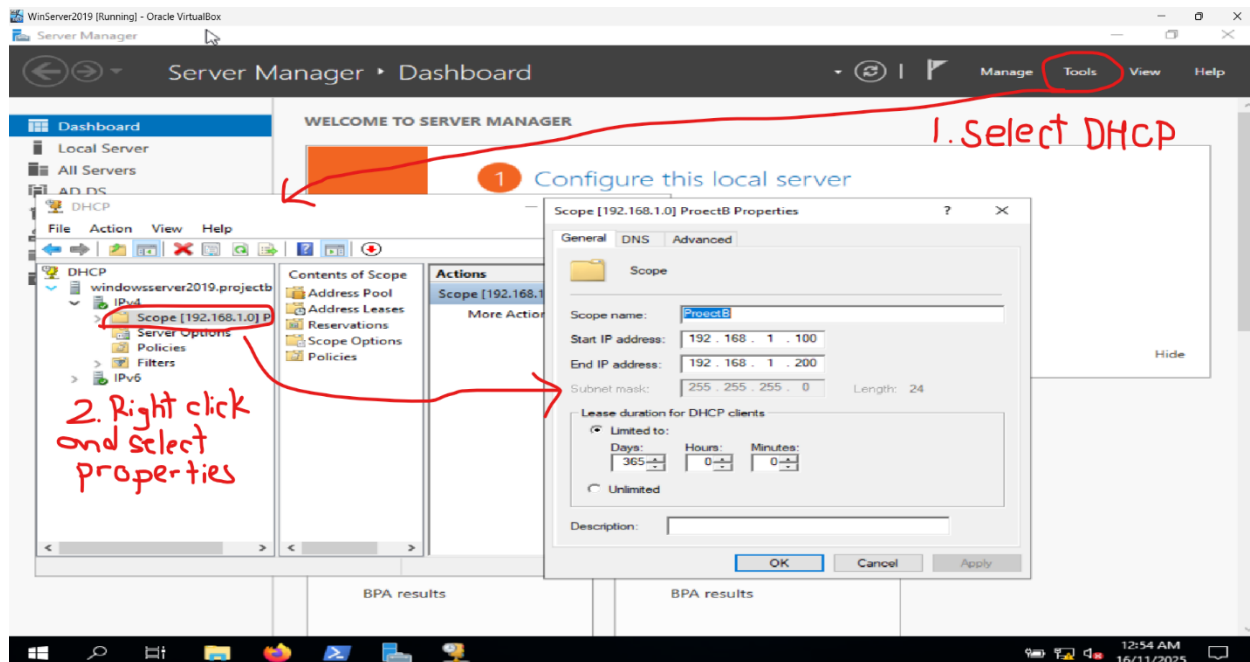
To assign the range of IP addresses that its DHCP can distribute, from Server Manager:

1. **Go to Tools > DHCP**
2. Expand **IPv4** and right click '**Scope**' and click '**Properties**'
3. In the '**Start IP address**', write starting IP address (**192.168.1.100**) and in the '**End IP address**', write ending IP address (**192.168.1.200**) that DHCP server can distribute and then click '**OK**'

**Figure 20**

*DHCP IP addressing configuration*





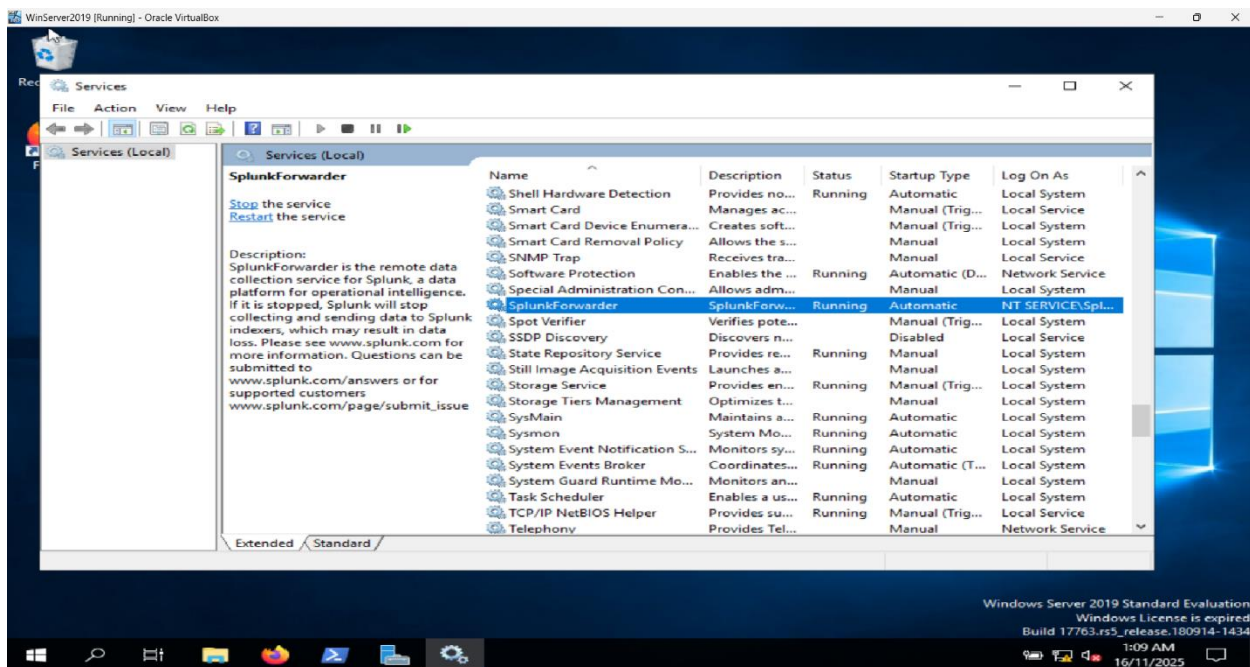
Note. This Figure demonstrates steps on how to set IP address range that are distributed

Windows Server cannot send logs to Splunk Enterprise on its own. It requires **the installation of Splunk Universal Forwarder**, another log forwarding SOC tool that collects and sends data. Download and run the installer from the Splunk home page and follow the instructions:

1. Enter the IP address of CentOS (**192.168.1.106**) and the management port (**8089**)
2. Set Username: forwarder and Password: forwarder
3. Enter the IP address of Splunk Enterprise server (**192.168.1.106**)
4. Enter the receiving port → **9997**
5. The installer will ask what kind of data to send. Check at least:  
**'Application Event Logs', 'System Event Logs', 'Security Event Logs' and 'IIS logs'**
6. Click **Install**. It will be installed on this directory: **'C:\Program Files\SplunkUniversalForwarder'**
7. After installation, the Splunk Universal Forwarder service will start automatically.
8. We can check it from **Services.msc** → look for **SplunkForwarder**.

**Figure 21**

## Splunk Universal Forwarder in Windows Server



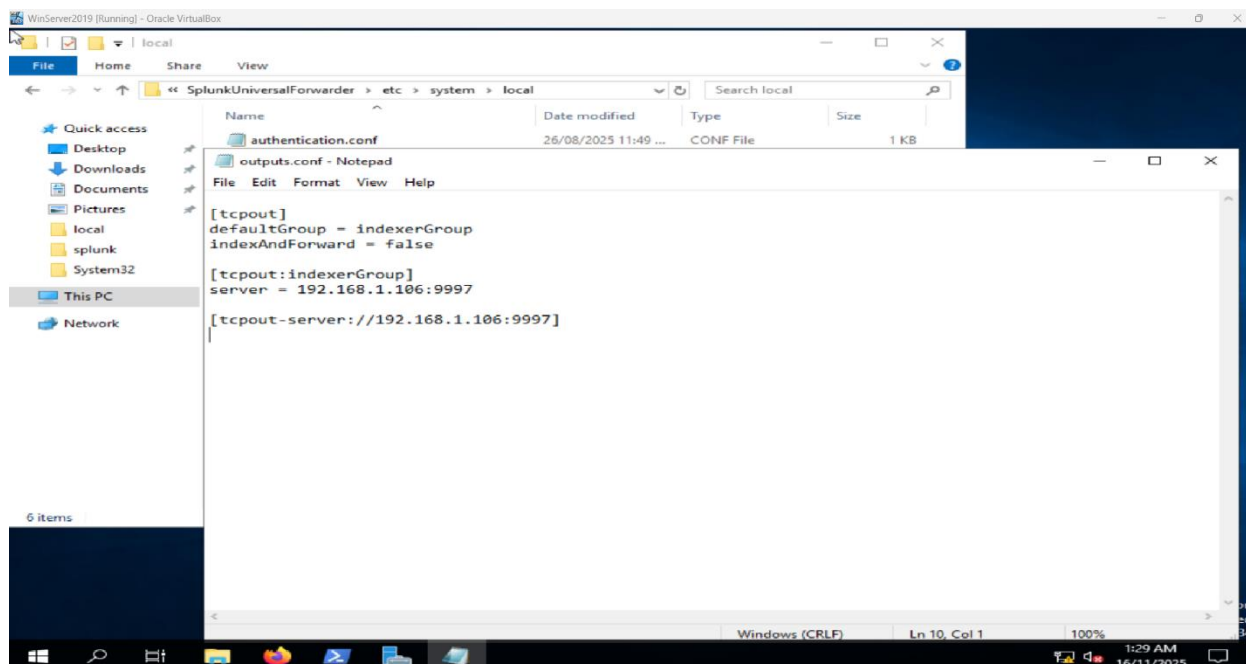
Note. This Figure demonstrates Splunk Universal Forwarder is successfully installed and running automatically in Windows Server

Once installation is complete, go to '**C:\Program**

**Files\SplunkUniversalForwarder\etc\system\local\**' and modify '**outputs.conf**' to reflect the IP address of the Splunk host (**192.168.1.106**) and port number **9997**:

**Figure 22**

*Outputs.conf configuration*



Note. This Figure demonstrates how 'outputs.conf' should look like

Once all these configurations are complete, Splunk Enterprise will be able to receive logs from Windows Server. To check logs in Splunk, refer to '**Figure 14**' in the '**CentOS and Splunk Enterprise**' section.

## Security Mechanisms

This section covers the simulation of security mechanisms in our SOC environment with tools from **Kali Linux**. The goal of these mechanisms is to test whether **PFSense Firewall** can detect malicious behaviour and protect our internal network from Kali's three attacks, while maintaining continuous visibility over network activity without disrupting business operations.

### Red Team

**Red Team** is a group of ethical hackers who perform cyberattack simulations to test how well an organisation's security controls can detect, resist, and respond to threats. This simulated attack helps an organisation identify vulnerabilities in its system and make improvements to its security operations (Anderson, Holdsworth, & Kosinski, n.d.). To simulate a Red Team environment, we performed three tests from Kali with various tools:

- **Network Service Discovery (T1046)** with **Nmap**

- **Brute Force: Password Guessing (T1110.001)** with **Hydra**
- **Active Scanning: Vulnerability Scanning (T1595.002)** with **Nikto**

‘**Network Service Discovery (T1046)**’ gets a listing of services running on remote hosts and local network infrastructure devices, including those that may be vulnerable to remote software exploitation (“Network Service Discovery”, 2017). In our test, we used ‘**nmap**’ to scan services running on Metasploitable by using this command from Kali:

**sudo nmap -T4 -Pn -A -v <Metasploitable’s IP>**

**Figure 23**

*Network Service Discovery simulation with Nmap*

```

kali@kali: ~
File Actions Edit View Help
(kali@kali)~$ sudo nmap -T4 -Pn -A -v 192.168.1.107
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-17 13:03 NZDT
NSE: Loaded 157 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 13:03
Completed NSE at 13:03, 0.00s elapsed
Initiating NSE at 13:03
Completed NSE at 13:03, 0.00s elapsed
Initiating NSE at 13:03
Completed NSE at 13:03, 0.00s elapsed
Initiating Parallel DNS resolution of 1 host. at 13:03
Completed Parallel DNS resolution of 1 host. at 13:04, 13.18s elapsed
Initiating SYN Stealth Scan at 13:04
Scanning 192.168.1.107 [1000 ports]
SYN Stealth Scan Timing: About 29.05% done; ETC: 13:05 (0:01:16 remaining)
SYN Stealth Scan Timing: About 58.70% done; ETC: 13:05 (0:00:43 remaining)
Completed SYN Stealth Scan at 13:05, 102.75s elapsed (1000 total ports)
Initiating Service scan at 13:05
Initiating OS detection (try #1) against 192.168.1.107
Retrying OS detection (try #2) against 192.168.1.107
Initiating Traceroute at 13:05
Completed Traceroute at 13:06, 9.08s elapsed
NSE: Script scanning 192.168.1.107.
Initiating NSE at 13:06
Completed NSE at 13:06, 5.02s elapsed
Initiating NSE at 13:06
Completed NSE at 13:06, 0.00s elapsed
Initiating NSE at 13:06
Completed NSE at 13:06, 0.00s elapsed

```

Note. This Figure demonstrates failed Network Service Discovery attack with Nmap

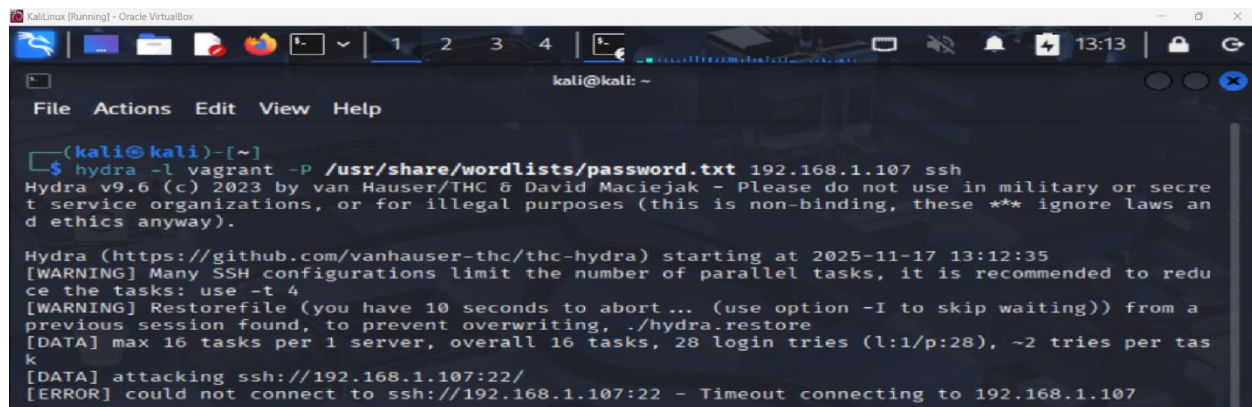
In our simulation, Metasploitable’s IP address is ‘**192.168.1.107**’. Once Nmap finishes scanning ports in Metasploitable, it lists all the open ports. Open services, such as FTP on port 21, SSH on port 22, Telnet on port 23, or HTTP on port 80, are the primary targets for the next Brute Force test.

**‘Brute Force: Password Guessing (T1110.001)’** is an attempt to guess the password of a target system using a repetitive or iterative mechanism (“Brute Force: Password Guessing”, 2020). We used **‘Hydra’** with a password list file called **‘password.txt’** to guess the credentials of Metasploitable:

**hydra -l vagrant -P /usr/share/wordlists/password.txt <Metasploitable’s IP> ssh**

**Figure 24**

*Brute Force simulation with Hydra*

A screenshot of a Kali Linux terminal window. The terminal shows the command `hydra -l vagrant -P /usr/share/wordlists/password.txt 192.168.1.107 ssh` being executed. The output shows Hydra v9.6 starting at 2025-11-17 13:12:35. It displays several warnings: "Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4", "Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.restore", and "[DATA] max 16 tasks per 1 server, overall 16 tasks, 28 login tries (l:1/p:28), ~2 tries per task". It then shows "[DATA] attacking ssh://192.168.1.107:22/" and finally "[ERROR] could not connect to ssh://192.168.1.107:22 - Timeout connecting to 192.168.1.107".

Note. This Figure demonstrates failed Brute Force attack with Hydra

After providing the username of Metasploitable (vagrant), Hydra goes through the list of passwords listed in ‘password.txt,’ and once it finds a match, it tells us the matching password.

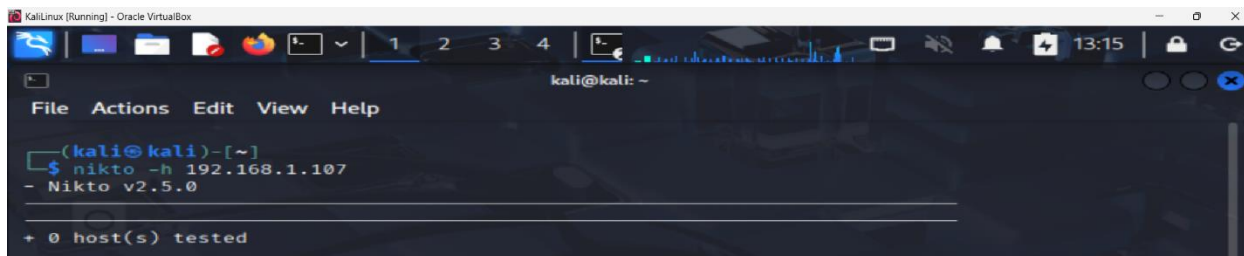
**‘Active Scanning: Vulnerability Scanning (T1595.002)’** checks the target’s vulnerabilities by scanning running software and version numbers, listening ports, or other network artifacts. Information from these may reveal opportunities for reconnaissance and initial access (“Active Scanning: Vulnerability Scanning”, 2020). We used **‘Nikto’** to scan the vulnerabilities of Metasploitable:

**nikto -h <Metasploitable’s IP>**

**Figure 25**

*Vulnerability Scanning with Nikto*





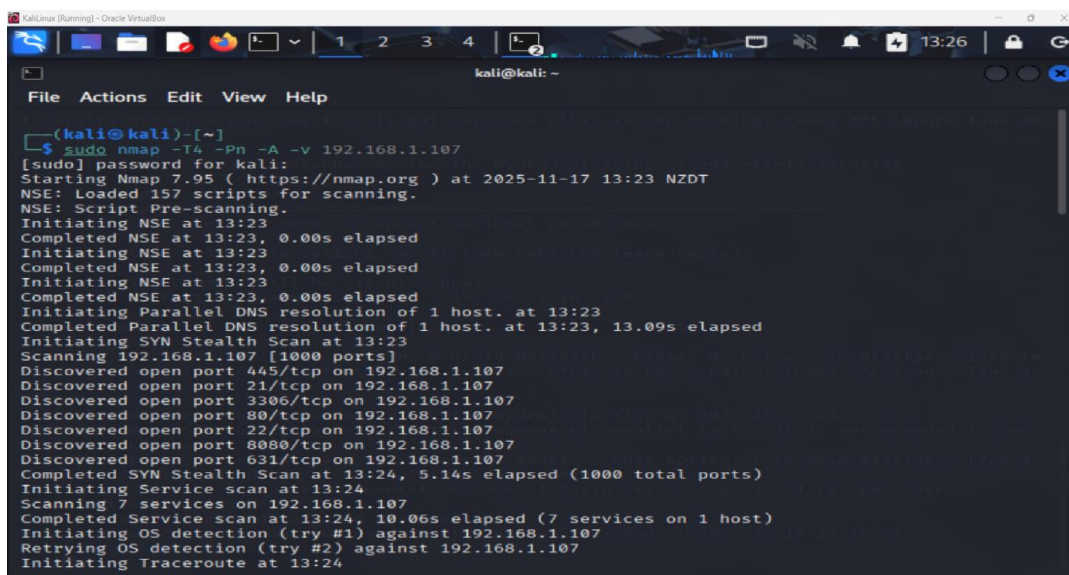
Note. This Figure demonstrates failed Vulnerability Scanning with Nikto

Once Nikto finishes scanning Metasploitable, it displays Metasploitable's IP address, hostname, port, server type, and other vulnerability issues, such as outdated software, exposed directories, and more.

By default, PFSense is configured to block all traffic from outside our local network, so all the tests, as shown in Figures 23, 24, and 25, failed. However, if PFSense is configured to allow Kali to communicate with Metasploitable, all the tests will be successful. This will be discussed in the '**Blue Team**' section. The Figures 26, 27, and 28 below show the results of successful tests:

**Figure 26**

*Successful Network Service Discovery simulation with Nmap*

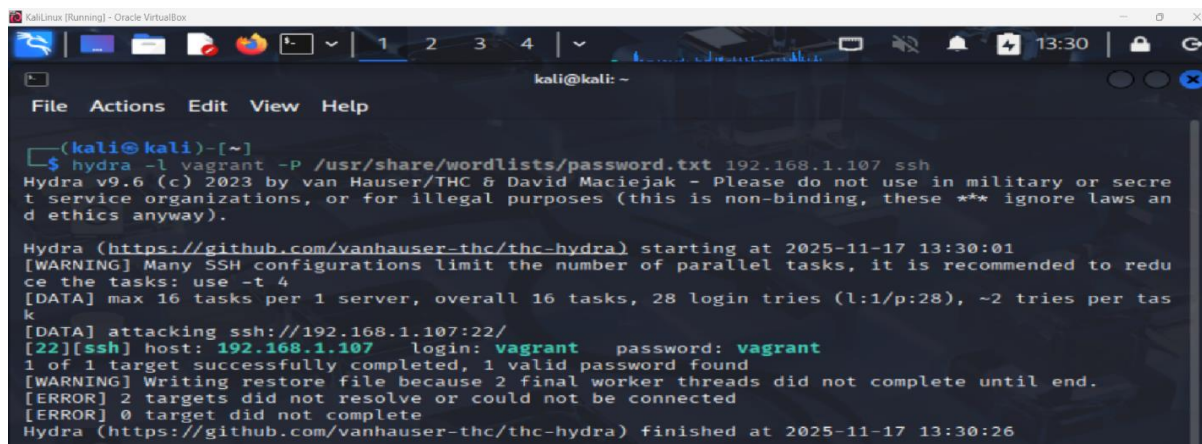


Note. This Figure shows seven open ports in Metasploitable

The successful network scanning from Figure 26 shows that nmap discovered seven open ports: 21, 22, 80, 445, 631, 3306, 8080 which indicates Metasploitable is vulnerable to Brute Force attack.

**Figure 27**

*Successful Brute Force simulation with Hydra*



```
(kali@kali)-[~]
$ hydra -l vagrant -P /usr/share/wordlists/password.txt 192.168.1.107 ssh
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

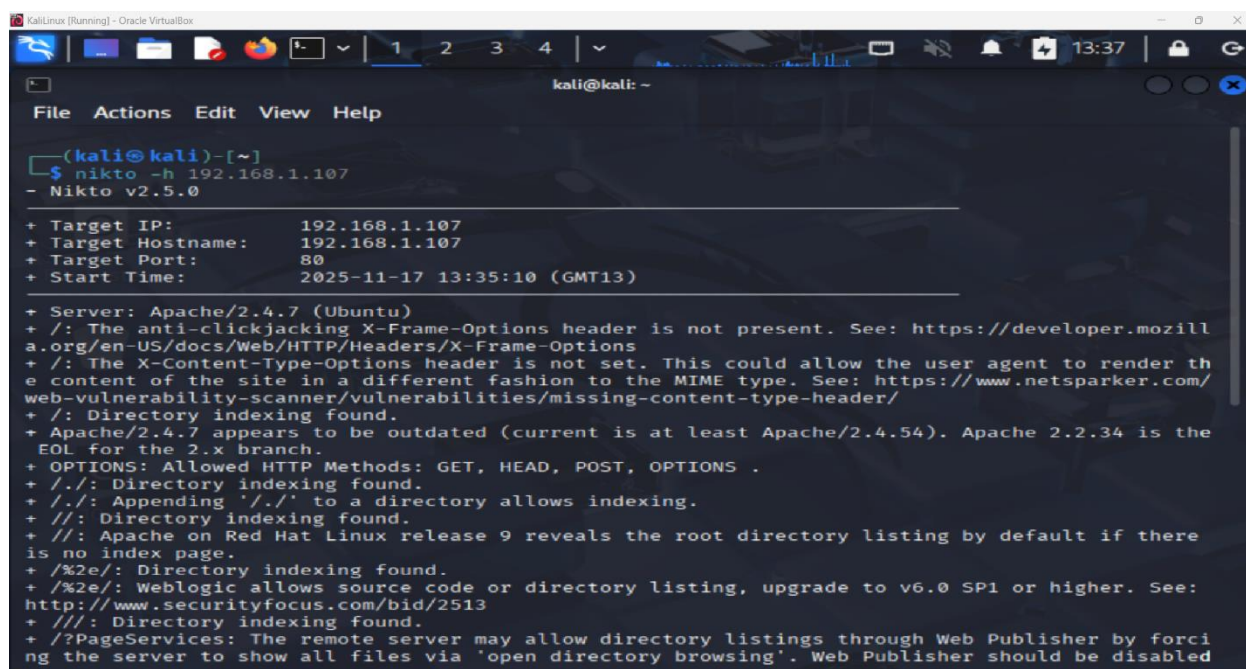
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-11-17 13:30:01
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 28 login tries (l:1/p:28), ~2 tries per task
[DATA] attacking ssh://192.168.1.107:22/
[22][ssh] host: 192.168.1.107 login: vagrant password: vagrant
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 2 final worker threads did not complete until end.
[ERROR] 2 targets did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-11-17 13:30:26
```

Note. This Figure shows the password of user ‘vagrant’ in Metasploitable

The successful Brute Force from Figure 27 shows that the user credential of ‘vagrant’ in Metasploitable is ‘vagrant’.

**Figure 28**

*Successful Vulnerability Scanning with Nikto*



```
(kali@kali)-[~]
$ nikto -h 192.168.1.107
- Nikto v2.5.0

+ Target IP: 192.168.1.107
+ Target Hostname: 192.168.1.107
+ Target Port: 80
+ Start Time: 2025-11-17 13:35:10 (GMT13)

+ Server: Apache/2.4.7 (Ubuntu)
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ /: Directory indexing found.
+ Apache/2.4.7 appears to be outdated (current is at least Apache/2.4.54). Apache 2.2.34 is the EOL for the 2.x branch.
+ OPTIONS: Allowed HTTP Methods: GET, HEAD, POST, OPTIONS .
+ /./: Directory indexing found.
+ /./: Appending '/./' to a directory allows indexing.
+ /./: Directory indexing found.
+ /./: Apache on Red Hat Linux release 9 reveals the root directory listing by default if there is no index page.
+ /%2e/: Directory indexing found.
+ /%2e/: Weblogic allows source code or directory listing, upgrade to v6.0 SP1 or higher. See: http://www.securityfocus.com/bid/2513
+ /./: Directory indexing found.
+ /?PageServices: The remote server may allow directory listings through Web Publisher by forcing the server to show all files via 'open directory browsing'. Web Publisher should be disabled
```

Note. This Figure shows existing vulnerabilities of Metasploitable





installed on CentOS (See **Figure 6** of ‘**PFSense Firewall**’ section). When the firewall detects any traffic against Metasploitable, it sends blocked/pass logs to Splunk:

**Figure 30**

### *Blocked logs from PFSense Firewall in Splunk Enterprise*

| i | Time                    | Event                                                                                                                                                                                                                                                 |
|---|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| > | 11/17/25 1:52:44.000 PM | Nov 17 13:52:44 192.168.1.1 Nov 17 13:52:44 filterlog[34351]: 60,,12001,em0,match,block,in,4,0x0,,41,44608,0,none,6,tcp,44,10.0.2.10,192.168.1.107,63250,9929,0,5,1498011693,,1024,,mss<br>host = 192.168.11   source = udp:514   sourcetype = syslog |
| > | 11/17/25 1:52:44.000 PM | Nov 17 13:52:44 192.168.1.1 Nov 17 13:52:44 filterlog[34351]: 60,,12001,em0,match,block,in,4,0x0,,41,44608,0,none,6,tcp,44,10.0.2.10,192.168.1.107,63250,9929,0,5,1498011693,,1024,,mss<br>host = 192.168.11   source = udp:514   sourcetype = syslog |
| > | 11/17/25 1:52:44.000 PM | Nov 17 13:52:44 192.168.1.1 Nov 17 13:52:44 filterlog[34351]: 60,,12001,em0,match,block,in,4,0x0,,54,16048,0,none,6,tcp,44,10.0.2.10,192.168.1.107,63250,5952,0,5,1498011693,,1024,,mss<br>host = 192.168.11   source = udp:514   sourcetype = syslog |
| > | 11/17/25 1:52:44.000 PM | Nov 17 13:52:44 192.168.1.1 Nov 17 13:52:44 filterlog[34351]: 60,,12001,em0,match,block,in,4,0x0,,54,16048,0,none,6,tcp,44,10.0.2.10,192.168.1.107,63250,5952,0,5,1498011693,,1024,,mss<br>host = 192.168.11   source = udp:514   sourcetype = syslog |
| > | 11/17/25 1:52:44.000 PM | Nov 17 13:52:44 192.168.1.1 Nov 17 13:52:44 filterlog[34351]: 60,,12001,em0,match,block,in,4,0x0,,47,25556,0,none,6,tcp,44,10.0.2.10,192.168.1.107,63250,2042,0,5,1498011693,,1024,,mss<br>host = 192.168.11   source = udp:514   sourcetype = syslog |
| > | 11/17/25 1:52:44.000 PM | Nov 17 13:52:44 192.168.1.1 Nov 17 13:52:44 filterlog[34351]: 60,,12001,em0,match,block,in,4,0x0,,47,25556,0,none,6,tcp,44,10.0.2.10,192.168.1.107,63250,2042,0,5,1498011693,,1024,,mss<br>host = 192.168.11   source = udp:514   sourcetype = syslog |
| > | 11/17/25 1:52:44.000 PM | Nov 17 13:52:44 192.168.1.1 Nov 17 13:52:44 filterlog[34351]: 60,,12001,em0,match,block,in,4,0x0,,40,63089,0,none,6,tcp,44,10.0.2.10,192.168.1.107,63250,5964,0,5,1498011693,,1024,,mss<br>host = 192.168.11   source = udp:514   sourcetype = syslog |
| > | 11/17/25 1:52:44.000 PM | Nov 17 13:52:44 192.168.1.1 Nov 17 13:52:44 filterlog[34351]: 60,,12001,em0,match,block,in,4,0x0,,40,63089,0,none,6,tcp,44,10.0.2.10,192.168.1.107,63250,5964,0,5,1498011693,,1024,,mss<br>host = 192.168.11   source = udp:514   sourcetype = syslog |

Note. This Figure demonstrates Splunk received list of blocked logs from PFSense

To see all the logs from PFSense, log into Splunk Enterprise, click ‘**Search & Reporting**’ and type the following in the search box:

**Index=\* host="192.168.1.1" "10.0.2.10" "192.168.1.107"**

This will list all the logs from PFSense that Splunk received. Logs from Figure 30 show that PFSense (**192.168.1.1**) blocked TCP traffic in port 44 from Kali (**10.0.2.10**) to Metasploitable (**192.168.1.107**).

To test the successful attack attempts and receive these logs, we need to configure Kali to communicate with Metasploitable from PFSense. From PFSense’s web interface:

1. Go to **Firewall > Rules > WAN** and click ‘**Add**’
2. In ‘**Edit Firewall Rule**’ section, set:
  - a. ‘**Action**’ = ‘**pass**’
  - b. ‘**Address Family**’ = ‘**IPv4**’

c. 'Protocol' = 'Any'

3. In 'Source' section, set 'Address or Alias' and write Kali's IP (10.0.2.10)
4. In 'Destination' section, set 'Address or Alias' and write Metasploitable's IP (192.168.1.107)
5. In 'Extra Options' section, tick 'Log packets that are handled by this rule'
6. Click 'Save'
7. Go to Interfaces > WAN
8. In 'Reserved Networks', untick 'Block private networks and loopback addresses' and 'Block bogon networks'
9. Click 'Save'

These configurations will allow attacks from Kali to Metasploitable to succeed:

Figure 31

Passed logs from PFSense Firewall in Splunk Enterprise

| Time                    | Event                                                                                                                                                                                                                                                                     |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11/17/25 2:00:40.000 PM | Nov 17 14:00:40 192.168.1.1 Nov 17 14:00:39 filterlog[34351]: 70,,,1762616806,em0,match,pass,in,4,0x0,,64,52645,0,D F,6,tcp,60,10.0.2.10,192.168.1.107,38362,21,0,S,1284615055,,64240,,mss;sackOK;TS;nop;wscale host = 192.168.1.1 source = udp:514 sourcetype = syslog   |
| 11/17/25 2:00:40.000 PM | Nov 17 14:00:40 192.168.1.1 Nov 17 14:00:39 filterlog[34351]: 70,,,1762616806,em0,match,pass,in,4,0x0,,64,52645,0,D F,6,tcp,60,10.0.2.10,192.168.1.107,38362,21,0,S,1284615055,,64240,,mss;sackOK;TS;nop;wscale host = 192.168.1.1 source = udp:514 sourcetype = syslog   |
| 11/17/25 2:00:32.000 PM | Nov 17 14:00:32 192.168.1.1 Nov 17 14:00:32 filterlog[34351]: 70,,,1762616806,em0,match,pass,in,4,0x0,,64,24838,0,D F,6,tcp,60,10.0.2.10,192.168.1.107,33478,21,0,S,3224362788,,64240,,mss;sackOK;TS;nop;wscale host = 192.168.1.1 source = udp:514 sourcetype = syslog   |
| 11/17/25 2:00:32.000 PM | Nov 17 14:00:32 192.168.1.1 Nov 17 14:00:32 filterlog[34351]: 70,,,1762616806,em0,match,pass,in,4,0x0,,64,24838,0,D F,6,tcp,60,10.0.2.10,192.168.1.107,33478,21,0,S,3224352788,,64240,,mss;sackOK;TS;nop;wscale host = 192.168.1.1 source = udp:514 sourcetype = syslog   |
| 11/17/25 2:00:25.000 PM | Nov 17 14:00:25 192.168.1.1 Nov 17 14:00:25 filterlog[34351]: 70,,,1762616806,em0,match,pass,in,4,0x0,,64,28664,0,D F,6,tcp,60,10.0.2.10,192.168.1.107,53244,3396,0,S,3460156817,,64240,,mss;sackOK;TS;nop;wscale host = 192.168.1.1 source = udp:514 sourcetype = syslog |
| 11/17/25 2:00:25.000 PM | Nov 17 14:00:25 192.168.1.1 Nov 17 14:00:25 filterlog[34351]: 70,,,1762616806,em0,match,pass,in,4,0x0,,64,28664,0,D F,6,tcp,60,10.0.2.10,192.168.1.107,53244,3396,0,S,3460156817,,64240,,mss;sackOK;TS;nop;wscale host = 192.168.1.1 source = udp:514 sourcetype = syslog |
| 11/17/25 2:00:24.000 PM | Nov 17 14:00:24 192.168.1.1 Nov 17 14:00:24 filterlog[34351]: 70,,,1762616806,em0,match,pass,in,4,0x0,,64,65494,0,D F,6,tcp,60,10.0.2.10,192.168.1.107,42348,21,0,S,161410194,,64240,,mss;sackOK;TS;nop;wscale host = 192.168.1.1 source = udp:514 sourcetype = syslog    |
| 11/17/25 2:00:24.000 PM | Nov 17 14:00:24 192.168.1.1 Nov 17 14:00:24 filterlog[34351]: 70,,,1762616806,em0,match,pass,in,4,0x0,,64,65494,0,D F,6,tcp,60,10.0.2.10,192.168.1.107,42348,21,0,S,161410194,,64240,,mss;sackOK;TS;nop;wscale host = 192.168.1.1 source = udp:514 sourcetype = syslog    |

Note. This Figure demonstrates Splunk received list of passed logs from PFSense

Logs from Figure 31 show that Kali (10.0.2.10) passed TCP traffic in port 60 to Metasploitable (192.168.1.107).

Our tests conclude that **the Blue Team** can successfully block all attacks against our local network, indicating that it is difficult for cyberattacks to bypass the firewall.



# Conclusion

---

Through this project, we discussed the fundamentals of SOC and successfully simulated it using a virtualised network composed of five integrated systems. The objective was to build, secure, test, and analyse an enterprise-style environment, and the outcomes demonstrate that all major SOC components operated as intended.

**SOC** is a team of security analysts responsible for monitoring, detecting, and responding to potential cybersecurity incidents in systems. Essential tools for SOC include **SIEM (Security Information and Event Management)**, **IDS / IPS (Intrusion Detection & Prevention) systems**, **Next-Generation Firewalls (NGFWs)**, **Endpoint Protection**, and **Vulnerability scanners**. To become a successful security analyst, not only do they need to possess skills in using these tools, but they also require **excellent communication skills** to work well in a team, **a broad knowledge of networks, operating systems, and threats**, and **strong analytical and problem-solving skills**.

The **PFSense Firewall** effectively controlled traffic between networks and generated logs for analysis. The **Windows Server 2019** machine provided essential infrastructure services, including DHCP, DNS, Active Directory, and IIS hosting. **Splunk Enterprise on CentOS** served as a central SIEM platform, capable of receiving and correlating logs from multiple sources.

**Red Team** tests using **Nmap**, **Hydra**, and **Nikto** from **Kali Linux** demonstrated how realistic attack behaviours appear within monitored environments. The **Blue Team's** defensive configurations, including firewall rules and log forwarding, enabled complete visibility of both blocked and successful attack attempts on **Metasploitable**. Log evidence in Splunk confirmed that the SOC could detect network scanning, brute force attempts, and vulnerability scans. These findings highlighted the importance of log collection, centralised visibility, and coordinated response in modern cybersecurity environments. Overall, the security mechanisms of our simulation were implemented correctly and met their technical objectives, providing a solid foundation for understanding real-world cybersecurity operations. Future improvements include integrating EDR tools, automating alerting through SOAR capabilities, or expanding the dataset with additional endpoints to simulate larger enterprise environments.

# Resources and References

---

## Tools

<https://app.diagrams.net/> for drawing network diagrams

<https://www.virtualbox.org/> for simulating tests in a virtual environment

<https://www.kali.org/> for Red Team Testing against the network

<https://www.pfsense.org/> for Blue Team testing

<https://www.centos.org/> for checking logs from PFSense Firewall and Windows Server 2019

<https://www.splunk.com/> for installation of Splunk Enterprise and Splunk Universal Forwarder

<https://splunkbase.splunk.com/app/1527> for installation of TA-Pfsense for log forwarding of PFSense to Splunk Enterprise

<https://www.microsoft.com/en-us/evalcenter/evaluate-windows-server-2019> for Windows Server 2019

<https://sourceforge.net/projects/metasploitable/> for vulnerability test

<https://sourceforge.net/projects/metasploitable3-ub1404upgraded/> for vulnerability test

## References

*Active Scanning: Vulnerability Scanning.* (2020, October 2). MITRE. Retrieved from <https://attack.mitre.org/techniques/T1595/002/>

Anderson, E., Holdsworth, J., & Kosinski, M. (n.d.). *What is red teaming?* IBM. Retrieved from <https://www.ibm.com/think/topics/red-teaming>

*Brute Force: Password Guessing.* (2020, February 11). MITRE. Retrieved from <https://attack.mitre.org/techniques/T1110/001/>

Cloudflare. (2025). *What is a next-generation firewall (NGFW)?* Retrieved from Learning Center: <https://www.cloudflare.com/learning/security/what-is-next-generation-firewall-ngfw/>

- Elastic. (2025). *Elastic fundamentals*. Retrieved from Docs: <https://www.elastic.co/docs/get-started>
- Finn, T., & Downie, A. (n.d.). *What is blue team?* IBM. Retrieved from <https://www.ibm.com/think/topics/blue-team>
- Kathryn Knerler, I. P. (2022). 11 Strategies of a World-Class Cybersecurity Operations Center. In I. P. Kathryn Knerler, *11 Strategies of a World-Class Cybersecurity Operations Center* (p. 452). Bedford, MA: The MITRE Corporation.
- Liquid Web. (2019, August). *What is CentOS? Everything You Need to Know*. Liquid Web. Retrieved from <https://www.liquidweb.com/blog/what-is-centos/>
- Microsoft. (2025). *What is endpoint detection and response (EDR)?* Retrieved from Microsoft Security: <https://www.microsoft.com/en-us/security/business/security-101/what-is-edr-endpoint-detection-response>
- Network Service Discovery. (2017, May 31). MITRE. Retrieved from <https://attack.mitre.org/techniques/T1046/>
- OWASP. (2025). *Vulnerability Scanning Tools*. Retrieved from OWASP: [https://owasp.org/www-community/Vulnerability\\_Scanning\\_Tools](https://owasp.org/www-community/Vulnerability_Scanning_Tools)
- Snort. (2025). *What is Snort?* Retrieved from Snort: <https://www.snort.org/>